

DATA SCIENCE PROJECT REPORT

E-Commerce Purchase Exploratory Data Analysis

Uncovering Customer Behavior, Demographics & Payment Trends

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Executive Summary

This report documents an Exploratory Data Analysis (EDA) conducted on a dataset of 10,000 simulated e-commerce transactions. The objective was to extract actionable insights about customer demographics, purchasing behavior, and payment preferences using Python-based data analysis and visualization techniques.

The analysis demonstrates proficiency in data wrangling, statistical summarization, customer segmentation, and visual storytelling — core competencies sought in data science and analytics roles.

Project Highlights

Metric	Detail
Dataset Size	10,000 rows × 14 columns
Data Quality	No null values; fully clean dataset
Average Purchase Value	\$50.35
Purchase Range	\$0.95 (min) — \$99.99 (max)

Metric	Detail
Top Demographics	Engineers (~11%), French speakers (~11%)
Tools Used	Python, Pandas, Seaborn, Matplotlib
Platform	Kaggle Notebook (Jupyter)

1. Introduction

1.1 Project Background

E-commerce analytics is a critical domain in modern business intelligence. Understanding who your customers are, when they buy, what they pay, and how they pay can drive smarter marketing strategies, personalized recommendations, and risk management decisions.

This project performs a structured EDA on a realistic e-commerce dataset sourced from Kaggle. It serves as a portfolio piece demonstrating the ability to ask the right business questions, derive insights from data, and communicate findings clearly.

1.2 Objectives

- Understand the structure and quality of the dataset
- Identify statistical properties of purchase prices
- Analyze customer demographics: job titles, languages, and email providers
- Uncover temporal patterns (AM vs PM purchasing behavior)
- Investigate payment methods and credit card usage
- Extract specific customer records via filtering and lookup operations

1.3 Business Questions Addressed

#	Business Question	Category
1	What is the average, highest, and lowest purchase price?	Pricing
2	Which job titles purchase most frequently?	Demographics
3	How many customers speak French (fr)?	Demographics
4	How many job titles contain 'Engineer'?	Demographics
5	Do AM or PM customers buy differently?	Behavior
6	How many Mastercard users spent above \$50?	Payment
7	How many credit cards expire in 2020?	Risk
8	What are the top 5 email domains?	Digital
9	Lookup by IP address and credit card number	Record Search

2. Dataset Overview

2.1 Data Source

The dataset was obtained from Kaggle and represents simulated e-commerce purchase records. It is commonly used in data science education for practicing EDA, pandas operations, and visualization.

2.2 Dataset Structure

The dataset contains 10,000 rows and 14 columns with no missing values, making it ideal for focused analysis without extensive preprocessing.

Column Name	Data Type	Description
Address	object	Customer's street address
Lot	object	Lot identifier code
AM or PM	object	Time of purchase (morning or afternoon/evening)
Browser Info	object	User agent / browser string
Company	object	Company associated with customer
Credit Card	int64	Credit card number (16-digit)
CC Exp Date	object	Credit card expiry (MM/YY format)
CC Security Code	int64	CVV/security code
CC Provider	object	Credit card provider (Visa, Mastercard, etc.)
Email	object	Customer email address
Job	object	Customer's job title
IP Address	object	Customer's IP address
Language	object	Preferred language (ISO code)
Purchase Price	float64	Transaction amount in USD

2.3 Data Quality Assessment

- Missing Values: 0 null values across all 14 columns — no imputation required
- Duplicate Records: Dataset treated as clean with unique transaction records
- Data Types: Numeric fields (Purchase Price, Credit Card, CC Security Code) are correctly typed; string fields are object type
- Credit Card Numbers: Stored as int64, appropriate for lookup operations

3. Methodology

3.1 Tools & Libraries

Library	Version/Role	Usage in Project
Python 3	Core language	All analysis and scripting
Pandas	Data manipulation	Loading, filtering, grouping, value_counts
Seaborn	Statistical visualization	histplot, countplot, boxplot
Matplotlib	Plot rendering	Figure display and customization
NumPy	Numerical support	Imported for array operations
Kaggle Notebook	Environment	Jupyter-based cloud execution

3.2 Analytical Approach

The analysis followed a structured EDA pipeline:

- Data Ingestion — Loading the CSV dataset using `pd.read_csv()`
- Initial Inspection — Checking head/tail rows, dtypes, shape, and null counts
- Univariate Analysis — Distributions and frequencies of individual columns
- Bivariate Analysis — Relationships between pairs (e.g., AM/PM vs Price)
- Filtering & Lookup — Extracting records based on specific conditions
- Visualization — Communicating insights through charts

3.3 Key Code Patterns Used

Data Loading & Inspection

```
- data = pd.read_csv('Ecommerce Purchases')
- data.info() # Shape, dtypes, null check
- data.isnull().sum() # Null values per column
```

Statistical Summary

```
- data['Purchase Price'].max() # 99.99
- data['Purchase Price'].min() # 0.95
- data['Purchase Price'].mean() # 50.35
```

Filtering & Conditional Queries

```
- data[data['Language'] == 'fr'] # French speakers
- data[data['Job'].str.contains('engineer', case=False)] # Engineers
- data[(data['CC Provider']=='Mastercard') & (data['Purchase Price']>50)]
```

Custom Functions

```
- # Count credit cards expiring in 2020
- for i in data['CC Exp Date']:
-     if i.split('/')[1] == '20': count += 1
```

Email Domain Extraction

```
- data['Email'].apply(lambda x: x.split('@')[1])
- data['temp'].value_counts().head(5)
```

4. Key Findings & Insights

4.1 Purchase Price Distribution

The purchase prices range from \$0.95 to \$99.99, with a fairly uniform distribution across the \$0–\$100 range. The mean purchase price is \$50.35, suggesting a balanced spread without extreme skew.

- Most purchases fall in the \$40–\$60 range (central cluster)
- The histogram with KDE curve confirms a near-uniform distribution
- No significant outliers — data represents a clean, synthetic dataset

Business Implication: A uniform price distribution suggests this dataset models a general marketplace rather than a niche or luxury segment.

4.2 Customer Demographics

Job Titles

Analysis of job titles reveals the top purchasing occupations. Engineers represent approximately 11% of all customers when combining all engineering sub-types, making them the single largest professional segment.

Rank	Job Category	Significance
1	Engineer (all types)	~11% of total customers; largest segment
2	Lawyers / Legal Professionals	High-value professional segment
3	Managers	Decision-makers with purchasing authority
4	Accountants / Analysts	Financially aware buyers
5	Teachers / Educators	Volume buyers

Language Demographics

French (fr) speakers number approximately 1,097 customers — roughly 11% of the dataset, consistent with the demographic diversity reflected across the 14+ language codes in the data.

Languages observed include English (en), French (fr), German (de), Spanish (es), Russian (ru), and others.

4.3 Time-Based Purchase Patterns

The dataset captures whether purchases were made in the AM or PM. The split is approximately equal, with a marginal PM majority:

Time Period	Purchase Count	Observation
AM (Morning)	~4,932	Slightly fewer transactions
PM (Afternoon/Evening)	~5,068	Marginally higher activity

The boxplot comparison of AM vs PM purchase prices shows near-identical medians and interquartile ranges, indicating that time of day does not significantly influence the amount spent. This suggests consistent purchasing behavior throughout the day.

4.4 Payment Analysis

Mastercard High-Value Purchases

A filtered query identified all customers who used Mastercard AND spent more than \$50. This segment represents potential high-value targets for premium marketing campaigns.

Credit Card Expiry Risk

A custom function scanned the CC Exp Date column and counted cards expiring in 2020 (year code '20'). This type of analysis is valuable for proactive customer outreach to update payment information and prevent failed transactions.

Top Email Providers

Rank	Email Provider	Business Use
1	hotmail.com	Largest segment — target for direct email campaigns
2	yahoo.com	Second-largest user base
3	gmail.com	Tech-savvy users
4	smith.com (and variants)	Custom/business domains
5	Other providers	Niche user bases

4.5 Record Lookup Capabilities

The analysis demonstrated practical data retrieval skills — a key competency for roles in data engineering and analytics:

- IP Address Lookup: Retrieved customer email for IP 132.207.160.22 → amymiller@morales-harrison.com
- Credit Card Lookup: Retrieved customer email for card number 4664825258997302

- These patterns simulate real-world CRM data queries and fraud investigation workflows

5. Visualizations

Three core visualizations were produced to communicate findings:

Visualization	Chart Type	Key Insight
Purchase Price Distribution	Histogram + KDE (Seaborn histplot)	Near-uniform spread from \$0–\$100; mean at \$50.35
Top 10 Job Titles by Purchase Count	Horizontal Bar Chart (countplot)	Engineers and legal/management roles dominate
AM vs PM Purchase Price Comparison	Box Plot (boxplot)	No significant price difference between time periods

All visualizations use Seaborn's built-in color palettes (Set1, Set2) for clear differentiation, with Matplotlib used for display rendering. The notebook includes inline plots for immediate interpretation.

6. Technical Skills Demonstrated

6.1 Python & Pandas Proficiency

- Loading and inspecting DataFrames (head, tail, info, dtypes, shape)
- Null value detection with isnull().sum()
- Statistical aggregation: max(), min(), mean(), value_counts()
- Boolean filtering with single and compound conditions
- String operations: str.contains() for pattern matching
- Custom Python functions for iterative column processing
- Feature engineering: extracting email domains into a new column

6.2 Data Visualization

- Seaborn histplot with KDE overlay for distribution analysis
- Seaborn countplot for categorical frequency visualization
- Seaborn boxplot for comparing distributions across groups
- Matplotlib plt.show() for rendering in Jupyter environments

6.3 Analytical Thinking

- Formulating business-relevant questions from raw data
- Structuring EDA into logical sections (Basics → Analysis → Insights)
- Identifying actionable segments (high-value Mastercard users, expiring cards)
- Cross-referencing records using lookup operations on key fields

7. Conclusion & Future Work

7.1 Summary

This EDA project demonstrates a complete data analysis workflow — from data loading and quality assessment through statistical analysis, visualization, and insight generation. The key findings include:

- Average purchase price of \$50.35 with near-uniform distribution
- Engineers are the most common purchasing job category (~11%)
- French speakers represent approximately 11% of the customer base
- AM and PM purchase behaviors are statistically similar in price
- Hotmail.com and Yahoo.com are the most common email providers
- Targeted segments identified: Mastercard high-value users, expiring 2020 cards

7.2 Potential Next Steps

Enhancement	Description	Technique
Customer Segmentation	Group customers by purchase behavior, language, and job	K-Means Clustering
Predictive Modeling	Predict purchase amount based on demographics	Linear Regression, Random Forest
Anomaly Detection	Flag suspicious transactions or unusual IP patterns	Isolation Forest
Time Series Analysis	Analyze purchase trends if timestamps are added	ARIMA, Plotly
Dashboard Creation	Interactive exploration of key metrics	Plotly Dash, Streamlit

7.3 Learning Outcomes

This project strengthened my ability to independently conduct end-to-end exploratory analysis, ask meaningful business questions from data, and present findings in a structured, portfolio-ready format. It reinforced the importance of data cleaning, feature understanding, and visual communication in real-world analytics work.

8. Project Information

Project Title	E-Commerce Purchase Exploratory Data Analysis
Author	Prince Maheta
GitHub Repository	github.com/princemaheta2627-glitch/ecommerce_purchase_eda
Platform	Kaggle Notebook (Jupyter)
Dataset	Ecommerce Purchases — Kaggle (10,000 rows, 14 columns)
Languages & Tools	Python 3, Pandas, Seaborn, Matplotlib, NumPy
Date Completed	June 2025
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This report was prepared as part of a data science portfolio.
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